

AUTONOMOUS PODCAST-BOOKING AGENT

Master Build Requirements – v2.0 (single source of truth)

Product in one sentence: User tells us *who* they're pitching (website, LinkedIn, client, or nothing) → the agent builds their profile, finds podcasts that *actually take guests*, finds host emails, writes personalized pitches (scored for quality), sends them (own inbox or our warmed pool), handles replies, books the interview, preps them, and shares the episode when it airs — **notifying by email at every step**, with **hard anti-spam guardrails** baked in — for one **flat monthly price**.

North-star promise: *"Give us a name or URL → watch it find your shows → get real results in your inbox in 60s → approve with one tap → it books, preps you, and shares the episode when it drops → guaranteed or we work free."*

1. Core flow

1. User provides **who to pitch** (any of: website, client site, LinkedIn, social, book, or nothing).
 2. Agent builds a **GuestProfile** (source-agnostic) + detects **existing appearances**.
 3. **~60s later**, a **results email** lands with real leads + one sample pitch → converts to signup.
 4. Agent queries **Podscan** → builds a ranked media list (topic fit + **guest-acceptance**).
 5. **Lazily**, only when a lead is approved to send, the agent runs the **waterfall email finder**.
 6. Agent writes **personalized, episode-referencing pitches** → each gets a **6-criteria quality score**.
 7. Agent **sends** (own inbox or pre-warmed pool with their display name) — **only if every guardrail passes**.
 8. **Email notifications** drive every step (leads ready → sent → replies → booked).
 9. Agent **monitors replies, classifies, auto-replies**, and **books** (calendar).
 10. **Post-booking loop:** prep-kit, episode-drop monitoring, auto share + thank-you.
 11. Reporting + flat monthly billing + reputation-based volume governance.
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2. Onboarding & Signup Funnel

2.1 The core insight: podcasts book *people*, not websites

The unit being pitched is a **person** (the guest), and a person's authority usually lives on **LinkedIn**, not a website. Agencies pitch many people. So onboarding must be **source-agnostic and per-client**, never "one website = one account."

Flexible input ("Add a person to pitch") — any combination, all optional:

Input	What it gives the agent
Website URL (own or client's)	Company value prop, product, proof, audience
Personal LinkedIn URL	The guest's authority: headline, about, experience, skills, articles, recommendations
LinkedIn company page	Company story when no good website exists
X/Twitter or Instagram handle	Voice, topics, reach, personality
Book / product / press-kit link	The asset + credibility (authors, speakers)
One free-text line	"I want to be known for X"
Nothing	→ fallback questions (§2.4)

The only requirement is *enough signal to build a profile* — and if the user gives nothing, we still proceed.

2.2 Source-agnostic profile builder (merge, don't replace)

- **LinkedIn = the person's authority** (title, bio, experience, skills, story).
- **Website = company/product + proof** (claims, numbers, customers).
- **Book/press = the asset + credibility.**
- **Social = voice + topics + reach.**
- Combine into one canonical `GuestProfile`.

`GuestProfile` **fields:** `identity` (name, title, company, headshot) · `bio` · `expertise_topics[]` · `authority_signals[]` (years, roles, books, awards, media) · `proof_points[]` · `talking_points[]` · `target_audience[]` · `media_appearances[]` · `podcast_ready_score` · `voice_samples[]`.

2.3 Existing-appearances detection (works even with *nothing but a name*)

- Query Podscan/web for the **person's name + company**; match episodes where they appear as a guest (reuse §5.1 guest-detection).
- Store: show, episode title, date, URL.
- Unlocks three things at once:
 1. **"Podcast-ready" proof** — "This person has been on 18 podcasts."
 2. **Seed targeting** — "You were on Show X → here are 20 like it" (best targeting signal that exists; no website needed).
 3. **Pitch credibility line** — "I've appeared on X and Y," inserted automatically.

2.4 Fallback (never block on missing inputs)

If inputs are too thin to extract expertise, **email 3 questions** (email-first, no dashboard): (1) Who are you and what do you do? (2) What 2–3 topics do you want to be known for? (3) Who's your ideal audience / what shows feel right? Agent parses the reply → builds profile.

2.5 The 60-second results email (highest-converting feature)

- After ~60s analysis, email **real results**: "For **[Name]**, we found **14 podcasts** that would book you" + **top 3 with fit scores** + one-line *why* + **one fully-written sample pitch** for the #1 match.
- The email *is* the demo. CTA: **"Approve & send these 3 free"** → free taste → upgrade.
- Urgency: *"Your matches refresh in 48h."*
- Track: delivered, opened, CTA clicked, signed up.

2.6 Other onboarding behaviors

- **Auto-suggest 3 talking points** from the profile; user clicks "looks right."
- **"Watch the agent work" theater** — live "Studying... mapping... scanning 3.8M shows... found 14 matches."
- **Login = magic link only**. No passwords. **No CC to start**.

2.7 Agency model (per-client profiles)

- Account = agency → holds **N client profiles** (roster). "Add client" = same flexible onboarding, nested under the agency.
- Each client = own inputs, `GuestProfile`, campaigns, voice, calendar, reporting, **billable seat**.
- Roster dashboard: per-client status (researching / pitching / replies / booked).
- Results email is **per-client**: "For **[Client Name]**, we found 14 shows."

3. Agent Orchestration

- Async pipeline: build_profile → detect_appearances → building_list → (lazy: finding_emails → writing → scoring → sending) → monitoring → booking → post_booking.
- Job queue; **progress emailed**, not just in-app.
- **Idempotent + resumable** — crash-safe, never double-send.
- **Daily run loop**: find N shows → draft → score → send (up to cap) → process replies → digest email.
- **Controls (email links or dashboard)**: daily cap (default ~20/day), pause/resume, exclude list, geography, min/max audience, avoid-topics.

4. Podcast Data — Podscan integration

- Use **Podscan API** for all podcast data. Do **not** build a crawler.
- Pull: title, description, category, host, RSS feed, recent episodes (title/description/date), **transcripts**.
- Podscan keyword/topic search + **alerts/mentions** to find shows discussing the subject.
- **Local snapshot cache** (avoid repeat lookups, unblock pitch gen).
- **Fit scoring (ours)**: topic overlap, recency, audience size, guest history, contact availability → 0–100.

5. Media List Building + Guest-Acceptance Signal

List build: dedupe (multi-feed), exclude inactive (>90 days), exclude already-contacted + DNC; saved lists; manual add/remove; learning loop feeds scoring.

5.1 Guest-Acceptance Signal (build first — #1 reply-rate lever)

AI detection from episode titles/descriptions/transcripts:

- Detect guest episodes ("joined by," "interview with," "featuring," "with [Name]," third-party intro).
 - Extract guest names/roles; classify (expert/author/founder/etc.).
 - Compute: **Takes Guests** (confident/likely/no) · **Guest cadence** (guests/mo) · **Guest recency** (days since last guest) · **Guest-type match** (do their guests resemble this person).
 - Surface: Takes guests:  (avg 4/mo, last guest 6d ago) + one-line evidence, on every lead card, in the results email, and in "why this show."
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6. Email Waterfall Finder (self-built, lazy, no Apollo)

Lazy/on-demand: run only when a lead is approved to send. Cache per podcast; re-verify stale (>30d) contacts. Biggest cost saver in the build.

Ordered cascade: (1) RSS feed owner email → (2) website `/contact / /about /footer mailto:` → (3) show notes "contact me" → (4) social bios → (5) pattern generation (`{first}@, {first}.` `{last}@, hello@, contact@, podcast@, host@`) → (6) WHOIS registrant email.

Verification (self-built): syntax + MX → SMTP RCPT TO → catch-all detection → disposable/role filter → store result + timestamp. Only "verified" or "high-confidence + catch-all" are sendable.

7. Pitch Generation + Live Quality Score

7.1 Generation (the quality moat)

- **Inputs:** profile, podcast metadata, **a specific recent episode** (title + transcript), host name, guest history.
- **Output (one unique email per show):** subject referencing the show/episode · opening referencing a **real episode/topic** (grounded, no hallucination) · 2–3 talking points · credibility line · one clear ask · plain text + light signature.
- **Voice matching:** upload samples → mimic tone; else clean default.
- **Hallucination guard:** every episode reference must exist in snapshot; validator rejects fabricated citations.

7.2 Live 6-criteria Quality Score (shown on the right side of the editor)

Every pitch gets a live **0–100 total**, computed from 6 weighted criteria. **The AI baseline is always ~100** — the score *drops live* as the user edits, showing exactly what broke and how to fix it.

#	Criterion	What it checks	Weight
1	Host / Show mention	Uses the host's or show's name	20%
2	Episode reference	Cites a real recent episode/topic	20%
3	Length	80–250 words	15%
4	Talking points	2–3 specific value points	15%

#	Criterion	What it checks	Weight
5	Credibility / proof	A "why me" proof point	15%
6	Clear ask + clean tone	One CTA; no ALL-CAPS, no !!!, ≤1 link	15%

Panel behavior:

- Each criterion shows a **pass/fail chip** + a fix hint ("Add the host's name," "Too long — cut to under 250 words").
- **Color bands:** 90+ green "Great" · 70–89 amber "Good" · <70 red "Needs work."
- **One-click "Restore AI version"** whenever the score drops.
- **Send gate:** score <70 blocks the send (ties into §21 guardrails).

8. Sending Infrastructure (dual mode)

Mode A — User's own inbox: Google OAuth (Gmail), Microsoft OAuth (Outlook), IMAP/SMTP (iCloud/Zoho). Encrypted tokens, refresh, disconnect/reconnect. Respect per-account daily caps.

Mode B — Our pre-warmed pool (hundreds of inboxes):

- Bulk-provision domains + mailboxes; auto SPF/DKIM/DMARC.
- **Warm-up engine:** ramp 5→20→50/day; monitor reputation.
- **User-set display name:** "From: **Jane Smith** <jane.smith@[warmed-domain]>". **Reply-To must route to the pooled inbox** (monitored → routed back), not the user's real email.
- Rotation + pinning (one inbox per thread); load-balance; never exceed per-inbox caps.
- Bounce/complaint handling (FBL), auto-suspend bad inboxes, auto-remove bounced contacts.
- Reply ingestion: IMAP polling/webhooks (Gmail push, Graph subscriptions).

9. Follow-ups

- Auto follow-ups as **"Re:" in the same thread**; schedule (default +3d, +7d), max 2.
- **Stop on reply — always.** Don't count against daily new-pitch cap.
- Each follow-up AI-drafted (new angle), shown in digest (or auto if opted in).

10. Reply Handling & Classification

- **Sync replies** into campaign inbox (thread view).
 - **Classify (LLM):** positive / negative / needs_more_info / out_of_office / technical / redirect / payroll / not_relevant.
 - **Extract:** requested materials, guest fee, location/format, **redirected/new email**, suggested times.
 - **Redirect capture:** "email my producer at X" → update contact + reroute.
 - **Auto-reply (optional):** positive / needs_more_info → answer + booking link; out_of_office → later follow-up; negative / payroll → polite close.
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11. Booking & Post-Booking Loop

11.1 True auto-booking: connect Calendly/Google Calendar → host says yes → send available times → book slot → email both parties a .ics → mark booked. (Default link if no calendar connected.)

11.2 Auto prep-kit (on booking): host one-pager (show summary, recent episodes, host socials, timezone, format) + tailored talking points.

11.3 Episode-drop monitoring (Podscan): detect when the episode airs → auto-thank-you to host → **auto-draft social posts** for the guest. Creates the "I got results" moment → renewals + referrals.

12. Email Notifications (first-class — email *is* the UX)

- **"Leads ready" email:** table of leads (show, host, fit score, guest-acceptance, subject, pitch preview) + **one-click Approve & Send / Edit / Skip** + bulk "Approve all (N)" / "Pause" + magic link.
 - **"Pitches sent" digest:** daily — sent, opens, replies, bookings.
 - **"New reply" alert:** reply text + **Approve AI reply / Write my own.**
 - **"Booked!"** notification + .ics.
 - **Weekly "results" email:** bookings + coaching ("these 3 shows replied — want 20 like them?").
 - **Billing:** receipt, payment-failed.
 - All action links = signed, single-use, time-boxed **magic links** (work logged-out).
 - **Preference center:** which emails, digest frequency, opt-out.
 - **Undo-send:** 5-minute undo window + preview on every send.
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13. Dashboard (supporting, not primary)

- Campaign home: status, weekly stats, pause/resume, daily cap.
 - Lead list: approve/edit/skip, sort by fit score **and guest-acceptance**, bulk actions.
 - **"Why this show"** (one-line reason) + **"More like this"** (find 20 like a winning lead).
 - Inbox: threads grouped by classification.
 - **Pitch editor with live 6-criteria score panel** (§7.2).
 - **Sender-health transparency**: live deliverability score + warning if user's inbox is risky.
 - Reports: per-campaign + aggregate; **unbranded CSV/Excel export**.
 - Account: connect email/calendar, voice samples, billing, team. **Team members free/unlimited**; roles owner/editor.
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14. Billing — flat monthly pricing + Performance Guarantee

- One plan: **single flat monthly price** (amount TBD). No per-email, per-lead, or seat charges.
 - 7-day free trial; monthly/quarterly; cancel anytime (pro-rated). Payment-failed → pause sending + email, keep data.
 - **Performance guarantee (headline next to price)**: *"4+ booked interviews in your first 30 days, or we keep working free until you get them."*
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15. Admin / Ops Panel

- Manage inbox pool (add domains/mailboxes, warm-up status, reputation, disable/rotate).
 - Deliverability metrics: bounce %, spam %, open/reply % per inbox/domain.
 - Global suppression + DNC ledger.
 - Usage audit per account; Podscan + LLM token spend.
 - **Content-moderation flag queue** (review flagged pitches/replies).
 - Kill-switch (platform-wide pause).
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16. 10/10 Feature Checklist (all required)

1. Guest-acceptance signal (§5.1)
2. Performance guarantee (§14)

3. Post-booking loop (§11)
4. Undo-send + 100% preview (§12)
5. Calendar integration → true auto-booking (§11.1)
6. "Watch the agent work" theater (§2.6)
7. "More like this" + "why this show" (§13)
8. Sender-health transparency (§13)
9. Host-side reverse marketplace (v2)
10. Agency/white-label mode (v2)

High-polish: URL/LinkedIn-only onboarding + auto-suggested talking points · magic-link everything · weekly results email with coaching · referral ("got booked? refer a founder, get a month free") · responsive dashboard (email is the mobile app) · **existing-appearances detection (§2.3)** · **live 6-criteria pitch score (§7.2).**

17. Data model (core entities)

- `users`, `workspaces`, `memberships` (role)
 - `client_profiles` (agency_id, guest profile fields, calendar_link, voice_samples, seat_status)
 - `guest_profiles` (identity, bio, expertise_topics[], authority_signals[], proof_points[], talking_points[], target_audience[], **media_appearances[]**, **podcast_ready_score**, voice_samples[])
 - `profile_sources[]` (which inputs provided + when)
 - `campaigns` (profile, settings, daily cap, mode, booking link)
 - `podcasts` (Podscan snapshot), `episodes` (title, date, transcript excerpt)
 - `guest_signals` (takes_guests, cadence, recency, guest_type_match, evidence)
 - `leads` (campaign + podcast + fit_score + guest_signal + status)
 - `contacts` (email candidates + source + confidence + verification + status)
 - `pitches` (lead + subject + body + episode_ref + **score_total** + **6 criteria scores** + status)
 - `sends` (pitch + inbox_id + message_id + delivered/bounced/opened)
 - `replies` (thread + classification + extracted fields)
 - `bookings` (lead + slot + calendar link + prep_kit_sent + aired)
 - `inboxes` (provider, email, domain, warm_up_state, reputation, status)
 - `suppression` (global DNC + cross-account recipient dedup ledger)
 - `reputation` (per account/domain/inbox: bounce%, complaint%, reply%)
 - `events` (notification history), `magic_links` (signed, expiring)
 - `subscriptions`, `invoices`
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18. Tech stack (recommendation — open to change)

The developer should propose the final stack; below is a starting point only. The requirements in this document are the source of truth.

- **Client:** React + TypeScript (or equivalent). Minimal UI — email is the primary UX.
 - **Server:** Node or Python. REST + light push/WebSocket.
 - **Queue:** durable job queue (Redis-based or similar) for pipeline, sends, warm-up, verification.
 - **DB:** PostgreSQL (or equivalent). **Cache:** Redis or similar. **Files:** S3-compatible.
 - **Email:** transactional ESP for notifications; **cold pitches via each inbox's SMTP/API** (never a shared marketing ESP).
 - **AI:** LLM for profiling, pitch writing, reply classification, guest detection, fit scoring, pitch scoring. Podscan for data. SMTP libraries for verification.
 - **Auth:** magic-link OTP + OAuth (Gmail/Outlook).
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19. Non-functional requirements

- **Deliverability is a top-line KPI:** track per-inbox bounce/spam/open/reply daily; auto-throttle.
 - **Compliance:** CAN-SPAM (physical address, opt-out), GDPR (erasure), global suppression + DNC.
 - **Security:** AES-256 at rest, TLS in transit, encrypted tokens, signed magic links, rate-limited auth.
 - **Never train on user data;** tenant isolation. **Audit log** of every send/approve/reply.
 - **SLO:** results email ~60s after input; sends within minutes; notifications <1 min.
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20. Milestones (4 phases)

M1 — Discover (the "find" half). Multi-input onboarding → profile builder + existing-appearances → 60s results email + theater → Podscan → list building + fit score → **guest-acceptance signal** → lazy waterfall finder + verification.

M2 — Write & Send (the "pitch" half). Pitch generation (grounding + guard) → **6-criteria live score panel** → sending infra (own inbox + warmed pool, display name, rotation, warm-up, undo-send) → follow-ups → email notifications + magic links + one-click approve → **re-validation gate + dynamic caps + suppression ledger**.

M3 — Close the loop. Replies (classification, redirect, auto-reply) → calendar auto-booking → prep-kit → episode-drop monitoring → weekly results email → reports/export.

M4 — Monetize & scale. Flat monthly billing + performance guarantee → admin/ops → team seats → v2: host reverse marketplace + agency/white-label.

21. Guardrails & Governance (the "don't let it become a spam cannon" system)

21.1 Core principle

Limits are deliverability-driven, not billing-driven — and enforced server-side, never trusted to the client. Flat pricing removes the cost brake, so the deliverability engine *is* the limiter.

21.2 Content guardrails (re-validation at send time)

Run on the **final edited text**, right before send — the user's edit is where spam enters (AI writes clean, the human breaks it).

Check	Rule	Action if failed
Pitch score	Total ≥ 70 (§7.2)	Block + show which criterion failed
Personalization floor	References specific show/host/episode	Block, offer AI version
Link hygiene	≤ 2 links, no shorteners/suspicious domains	Block
Formatting	No ALL-CAPS, no !!!, no pressure words	Flag
Near-duplicate	Not >90% identical across leads	Block + warn
Length	80–250 words	Warn/block

Rule: block + explain, never silently rewrite. Offer one-click "use the AI version." The personalization floor is a *selling point*: "We never let spray-and-pray leave the platform."

21.3 Volume guardrails (four levels, all must pass)

- **Per-campaign:** default ~20/day, user-settable down only.

- **Per-inbox:** Gmail ~40–100/day; Outlook/iCloud/Zoho conservative; pool inboxes 30–50/day.
- **Per-domain:** aggregate across all inboxes on a domain (e.g., 150/day).
- **Per-account:** aggregate across all campaigns + client profiles — no ceiling bypass via more inboxes/clients.

Plus: rate limiting (max/hour, minimum send gap) + business-hours sending windows in recipient timezone.

21.4 Dynamic, reputation-based caps

- **Master metric = spam-complaint rate** (<0.1% target, 0.3% hard max). Secondary: bounce <2%, positive-reply rate.
- New account starts at floor; **healthy** → **auto-raise** weekly (20→30→50→80); **deteriorating** → **auto-throttle**, then **pause + email why**; **hard breach** → **instant pause + admin review**.

21.5 Suppression & recipient protection

- **Cross-account recipient dedup:** same podcast pitched only once per platform per N months (shared ledger) — protects hosts from being swarmed by multiple agency clients.
- **Auto-suppress on opt-out language** in replies ("unsubscribe," "remove me") — permanent + global.
- **DNC list, role-address filter** (noreply@, abuse@, support@), **frequency cap** (once/host per X months), **one-email-per-thread**.

21.6 Infrastructure protection

- **Quality-tier isolation:** risky users send on separate domains from proven users.
- **Pool rotation** on degraded domains; **FBL processing** (Google/Yahoo) → auto-suppress + suspend; **auto-suspend inboxes/accounts** on breach; **global kill-switch**; **weekend/holiday send block** (configurable).

21.7 Compliance floor

CAN-SPAM (physical address + opt-out) · GDPR erasure · no purchased lists · real reachable sender identity on every send.

21.8 User-facing feedback (restrictions as trust)

- **Sender-health score** on dashboard (backed by real bounce/complaint/reply metrics).
- **"Why you're capped"** email — exact metric + one fix.

- **Positive reinforcement** — "Reply quality looks great — auto-raised your cap to 30."
- Every blocked pitch offers "**use the AI version**" as one-click recovery.